

Evidence Boundary Architecture

AgenThink Mesh Pharma Validation Library

Institutional-Grade Governance Framework for Retrospective Phase II → Phase III Validation

Classification: Governance Document — Internal & External Review

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EXECUTIVE VERDICT (Opinion First)

Would this architecture satisfy a skeptical FDA reviewer who believes the Council may have known the answer?

PARTIALLY — with a clear path to YES.

The architecture described in this document, if fully implemented, would satisfy the FDA's reproducibility and audit trail requirements. It would satisfy the EMA's methodology transparency requirements. It would satisfy Big Four chain-of-custody requirements. It would not fully satisfy the FDA's requirement for verifiable LLM training data exclusion — because that problem cannot be solved by architecture alone. It requires a fundamental change to the council execution model: replacing LLM-native knowledge with a document-retrieval-only architecture (Retrieval-Augmented Generation with a verified, date-bounded corpus).

The minimum viable version described in Part 10 is implementable in 60–90 days. The enterprise version requires 6–12 months. The regulator-ready version requires 18–24 months and a formal FDA/EMA pre-submission meeting.

The single most important action: Build the Evidence Corpus before running any more cases. Every case run without a verified evidence corpus is a case that cannot be defended.

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PART 1 — Evidence Boundary Framework

The Fundamental Problem

The Evidence Boundary Framework exists to answer one question: **“How do we know the Council did not already know the outcome?”**

This question has two distinct components. The first is architectural: was the council's input restricted to pre-decision evidence? The second is epistemological: was the council's underlying model trained on post-decision information? The architecture can fully solve the first. It can partially mitigate the second. It cannot eliminate the second without replacing the LLM execution model entirely.

This framework addresses both components with different tools and different honesty about

what each tool achieves.

Evidence Admissibility Standard

Category A — Admissible Evidence (Pre-Decision Corpus)

All evidence in this category is admissible for council input, provided it was publicly available before the Evidence Cutoff Date (defined in Part 2).

Evidence Type	Admissibility Condition	Verification Method
Phase I clinical trial results	Published or registered before cutoff	PubMed DOI + ClinicalTrials.gov NCT registration date
Phase II clinical trial results	Published or registered before cutoff	PubMed DOI + ClinicalTrials.gov NCT registration date
Investigator Brochure (IB)	Version dated before cutoff (if public)	FDA/EMA public disclosure or company press release with date
IND/CTA filing data	Only if publicly disclosed before cutoff	FDA public disclosure database; EMA clinical data portal
Preclinical data	Published before cutoff	PubMed DOI
Regulatory guidance documents	Issued before cutoff	FDA.gov / EMA.europa.eu with issue date
Published safety data	Published before cutoff	PubMed DOI; MedWatch if public
Competitive landscape	Published market analyses dated before cutoff	Source URL + publication date
Scientific literature	Published before cutoff	PubMed DOI; EMBASE; Cochrane
Conference abstracts	Presented before cutoff	Conference proceedings with date; PubMed abstract
ClinicalTrials.gov registry entries	Posted before cutoff	NCT registration date (verifiable)
FDA Advisory Committee transcripts	Published before cutoff	FDA.gov with meeting date
EMA Scientific Advice	Public disclosure before cutoff	EMA.europa.eu
Analyst reports	Published before cutoff	Source URL + publication date
Patent filings	Filed before cutoff	USPTO/EPO with filing date

Category B — Prohibited Evidence (Post-Decision Corpus)

All evidence in this category is prohibited from council input. Inclusion of any Category B evidence constitutes an Evidence Boundary Violation (EBV) and requires case invalidation.

Evidence Type	Prohibition Reason	Detection Method
Phase III trial results	Post-decision outcome	Publication date check
FDA approval/rejection decisions	Post-decision regulatory outcome	FDA.gov decision date
EMA approval/rejection decisions	Post-decision regulatory outcome	EMA.europa.eu decision date
MHRA decisions	Post-decision regulatory outcome	MHRA.gov.uk decision date
Post-decision press releases	Outcome disclosure	Company press release date
Market performance data	Revenue outcome	Financial report date
Post-decision analyst commentary	Outcome-informed analysis	Publication date check
Retrospective review articles	Outcome-informed synthesis	Publication date check
Post-decision safety updates	Outcome-informed safety data	Publication date check
DSMB interim analysis results	Outcome-informed safety data	Unless publicly disclosed before cutoff
Post-decision academic commentary	Outcome-informed analysis	Publication date check
Wikipedia articles	May contain post-decision information	Excluded entirely — not a primary source
News articles	May contain post-decision information	Date-checked; excluded if post-cutoff
LLM-generated summaries	May contain post-decision information	Excluded entirely — not a primary source

Category C — Conditional Evidence (Requires Review)

Evidence in this category requires case-by-case review by the Evidence Curator before admission.

Evidence Type	Condition for Admission	Review Required
Conference abstracts	Must be from conference held before cutoff	Evidence Curator review
Preprints (bioRxiv, medRxiv)	Must be posted before cutoff; note preprint status	Evidence Curator review
FDA correspondence	Only if publicly disclosed before cutoff	Evidence Curator review
Unpublished company data	Only if disclosed in public forum before cutoff	Evidence Curator review
Systematic reviews	Must be published before cutoff; check included studies	Evidence Curator review
Meta-analyses	Must be published before cutoff; check included studies	Evidence Curator review

The Formal Admissibility Standard

AgenThink Mesh Evidence Admissibility Standard v1.0

A document or data element is admissible as council input if and only if:

- (a) It is a primary source (peer-reviewed publication, regulatory document, clinical trial registry entry, or company press release); AND
- (b) It was publicly available before the Evidence Cutoff Date for the case; AND
- © It has been assigned a Document ID in the Evidence Manifest for the case; AND
- (d) Its retrieval date has been recorded; AND
- (e) It has been reviewed and approved by the Evidence Curator.

Any document that fails conditions (a)–(e) is inadmissible. Inclusion of an inadmissible document constitutes an Evidence Boundary Violation requiring case review.

PART 2 — Time Cutoff Governance

Decision Date Definition

The **Decision Date** is the date on which the sponsor made the formal internal decision to advance the drug from Phase II to Phase III. This is the date that defines the evidence boundary.

Establishing the Decision Date — Priority Order:

1. **Board or executive committee minutes** (if publicly disclosed) — highest reliability
2. **Phase III IND/CTA filing date** — the IND/CTA cannot be filed without a prior advancement decision; filing date is a reliable upper bound
3. **Phase III ClinicalTrials.gov registration date** — NCT registration is required before first patient enrolment; registration date is a reliable upper bound
4. **First patient enrolled date** — lower bound; advancement decision preceded this
5. **Company press release announcing Phase III** — reliable if dated; may lag the internal decision by weeks
6. **Phase III protocol publication** — reliable if dated

Decision Date Rule: When multiple sources are available, the earliest verifiable date is used as the Decision Date. This is the most conservative approach — it minimises the risk of including post-decision evidence.

Decision Date Uncertainty: When the Decision Date cannot be established within a 90-day window, the case is flagged as **Decision Date Uncertain (DDU)**. DDU cases are admitted to the library with a disclosed uncertainty range and are excluded from sensitivity/specificity calculations until the date is resolved.

Evidence Cutoff Date

The **Evidence Cutoff Date** is defined as: **Decision Date minus 30 days**.

The 30-day buffer accounts for:

- Publication lag between data availability and formal publication
- Internal review periods before public disclosure
- The realistic information environment of a decision-maker at the time

Rationale for 30-day buffer: A decision-maker advancing a drug to Phase III in January 2006 would not have had access to a paper published in December 2005 if it appeared online in late December. The 30-day buffer is conservative and defensible.

Evidence Cutoff Date = Decision Date – 30 days

Publication Cutoff Rules

Source Type	Cutoff Rule	Verification
Peer-reviewed journals	Online publication date (Epub ahead of print)	PubMed “Epub” date field
Conference abstracts	Abstract submission date OR conference date, whichever is earlier	Conference proceedings
ClinicalTrials.gov	Record posting date	NCT record “First Posted” field
Analyst reports	Report publication date	Source URL + date stamp
Company press releases	Press release date	Company website + PR Newswire
FDA documents	Document publication date	FDA.gov
EMA documents	Document publication date	EMA.europa.eu
Preprints	Preprint posting date	bioRxiv/medRxiv “Posted” date
Patents	Filing date (not grant date)	USPTO/EPO filing date

Governance Rules — Edge Cases

Rule 2.1 — Embargo Violations: If a paper was published after the Evidence Cutoff Date but was presented at a conference before the cutoff, the conference abstract (not the full paper) is admissible. The full paper is not admissible.

Rule 2.2 — Retrospective Registrations: ClinicalTrials.gov registrations that were posted after the trial started (retrospective registrations) are admissible only if the registration date is before the Evidence Cutoff Date. The registration date, not the trial start date, governs admissibility.

Rule 2.3 — Rolling Publications: For papers published in parts (e.g., interim analyses), each publication is evaluated independently. The first publication is admissible if before the cutoff; subsequent publications are evaluated by their own publication dates.

Rule 2.4 — Corrected Publications: If a paper was corrected after the Evidence Cutoff Date, the original version (as published before the cutoff) is admissible. The correction is not admissible.

Rule 2.5 — Withdrawn Publications: If a paper was withdrawn after the Evidence Cutoff Date, the original version is admissible if it was published before the cutoff. The withdrawal notice is not admissible.

Rule 2.6 — Analyst Reports: Analyst reports are admissible as Category A evidence only if they are from a named analyst at a named firm with a verifiable publication date. Anonymous or undated analyst commentary is inadmissible.

Rule 2.7 — Wikipedia: Wikipedia is never admissible as a primary source. It may be used to identify primary sources, which are then independently verified.

PART 3 — Evidence Manifest System

Evidence Manifest Purpose

The Evidence Manifest is the chain-of-custody document for each case. It records every document considered for council input, whether admitted or excluded, with the rationale for each decision. An independent reviewer must be able to reconstruct the council's evidence package from the manifest alone.

Evidence Manifest Schema

Case-Level Fields

Field	Type	Description	Required
case_id	String	Unique case identifier (format: ATM-PHARMA-[YEAR]-[SEQ])	Yes
drug_name	String	Generic drug name	Yes
brand_name	String	Brand name (if approved) or “N/A”	Yes
sponsor	String	Sponsor company at time of decision	Yes
indication	String	Specific disease/condition	Yes
therapeutic_area	String	Therapeutic area (controlled vocabulary)	Yes
decision_date	Date	ISO 8601 format (YYYY-MM-DD)	Yes
decision_date_source	String	Source used to establish decision date	Yes
decision_date_confidence	Enum	HIGH / MEDIUM / LOW / DDU	Yes
evidence_cutoff_date	Date	Decision Date – 30 days (ISO 8601)	Yes
manifest_version	String	Semantic version (e.g., “1.0.0”)	Yes
manifest_created_date	Date	ISO 8601	Yes
manifest_created_by	String	Evidence Curator name and role	Yes

Field	Type	Description	Required
manifest_reviewed_by	String	Independent reviewer name and role	Yes
manifest_approved_date	Date	ISO 8601	Yes
constitution_version	String	Pharma Constitution version used	Yes
council_run_date	Date	ISO 8601	Yes
council_session_id	String	Unique session identifier	Yes
total_documents_considered	Integer	Total documents reviewed for admission	Yes
total_documents_admitted	Integer	Documents admitted to evidence corpus	Yes
total_documents_excluded	Integer	Documents excluded with rationale	Yes
evidence_boundary_violations	Integer	Count of EBVs detected and resolved	Yes
case_status	Enum	ACTIVE / SUSPENDED / INVALIDATED	Yes

Document-Level Fields (per admitted document)

Field	Type	Description	Required
<code>doc_id</code>	String	Unique document identifier (format: DOC-[CASE_ID]-[SEQ])	Yes
<code>doc_type</code>	Enum	JOURNAL_ARTICLE / CONFERENCE_ABSTRACT / REGULATORY_DOCUMENT / CLINICAL_TRIAL_REGISTRY / PRESS_RELEASE / ANALYST_REPORT / PATENT / OTHER	Yes
<code>title</code>	String	Full document title	Yes
<code>authors</code>	String	Author list (last name, first initial)	Yes
<code>publication_name</code>	String	Journal, conference, or source name	Yes
<code>publication_date</code>	Date	ISO 8601 (online publication date for journals)	Yes
<code>doi_or_url</code>	String	DOI (preferred) or URL	Yes
<code>retrieval_date</code>	Date	Date document was retrieved by Evidence Curator	Yes
<code>admissibility_category</code>	Enum	A / B / C	Yes
<code>admission_decision</code>	Enum	ADMITTED / EXCLUDED	Yes
<code>exclusion_reason</code>	String	Required if EXCLUDED; must reference specific rule	Conditional
<code>evidence_type</code>	String	What the document provides (e.g., "Phase II efficacy data")	Yes

Field	Type	Description	Required
key_data_points	String	Specific data points extracted for council input	Yes
leakage_risk_flag	Boolean	True if document required leakage review	Yes
leakage_review_notes	String	Required if leakage_risk_flag is True	Conditional
curator_notes	String	Any additional notes from Evidence Curator	No

Exclusion Log Fields (per excluded document)

Field	Type	Description	Required
doc_id	String	Unique document identifier	Yes
title	String	Full document title	Yes
publication_date	Date	ISO 8601	Yes
doi_or_url	String	DOI or URL	Yes
exclusion_category	Enum	POST_CUTOFF / OUTCOME_CONTAMINATION / NOT_PRIMARY_SOURCE / INSUFFICIENT_DATE_VERIFICATION / OTHER	Yes
exclusion_rule	String	Specific admissibility rule violated	Yes
curator_decision	String	Evidence Curator's decision rationale	Yes

Sample Evidence Manifest Entry — Torcetrapib Case

```
{
  "case_id": "ATM-PHARMA-2006-001",
  "drug_name": "torcetrapib",
  "brand_name": "N/A (never approved)",
  "sponsor": "Pfizer Inc.",
  "indication": "Dyslipidaemia / cardiovascular risk reduction",
  "therapeutic_area": "Cardiovascular",
  "decision_date": "2004-10-01",
  "decision_date_source": "Phase III ClinicalTrials.gov registration NCT00101429",
  "decision_date_confidence": "MEDIUM",
  "evidence_cutoff_date": "2004-09-01",
  "manifest_version": "1.0.0",
  "manifest_created_date": "2026-06-14",
  "manifest_created_by": "Evidence Curator — AgenThink Mesh",
  "manifest_reviewed_by": "Independent Pharmaceutical Expert — [Name]",
  "manifest_approved_date": "2026-06-14",
  "constitution_version": "Pharma Constitution V1.0",
  "council_run_date": "2026-06-11",
  "council_session_id": "PHARMA-RETRO-TORCETRAPIB-1781180831427",
  "total_documents_considered": 12,
  "total_documents_admitted": 8,
  "total_documents_excluded": 4,
  "evidence_boundary_violations": 0,
  "case_status": "ACTIVE",
  "admitted_documents": [
    {
      "doc_id": "DOC-ATM-2006-001-001",
      "doc_type": "JOURNAL_ARTICLE",
      "title": "Effects of torcetrapib on HDL cholesterol and atherosclerosis in patients with type 2 diabetes mellitus: a randomised controlled trial",
      "authors": "Brousseau ME, Schaefer EJ, Wolfe ML, et al.",
      "publication_name": "New England Journal of Medicine",
      "publication_date": "2004-06-03",
      "doi_or_url": "10.1056/NEJMoa031766",
      "retrieval_date": "2026-06-10",
      "admissibility_category": "A",
      "admission_decision": "ADMITTED",
      "evidence_type": "Phase II efficacy data — HDL-C increase, LDL-C reduction",
      "key_data_points": "HDL-C increase 46-106% dose-dependent; LDL-C reduction 50-65% dose-dependent",
      "leakage_risk_flag": false
    },
    {
      "doc_id": "DOC-ATM-2006-001-002",
      "doc_type": "JOURNAL_ARTICLE",
      "title": "Effects of torcetrapib on HDL cholesterol and atherosclerosis in patients with type 2 diabetes mellitus: a randomised controlled trial",
      "authors": "Brousseau ME, Schaefer EJ, Wolfe ML, et al.",
      "publication_name": "New England Journal of Medicine",
      "publication_date": "2004-06-03",
      "doi_or_url": "10.1056/NEJMoa031766",
      "retrieval_date": "2026-06-10",
      "admissibility_category": "A",
      "admission_decision": "ADMITTED",
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      "leakage_risk_flag": false
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      "doc_type": "JOURNAL_ARTICLE",
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      "authors": "Brousseau ME, Schaefer EJ, Wolfe ML, et al.",
      "publication_name": "New England Journal of Medicine",
      "publication_date": "2004-06-03",
      "doi_or_url": "10.1056/NEJMoa031766",
      "retrieval_date": "2026-06-10",
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      "admission_decision": "ADMITTED",
      "evidence_type": "Phase II efficacy data — HDL-C increase, LDL-C reduction",
      "key_data_points": "HDL-C increase 46-106% dose-dependent; LDL-C reduction 50-65% dose-dependent",
      "leakage_risk_flag": false
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      "doc_type": "JOURNAL_ARTICLE",
      "title": "Effects of torcetrapib on HDL cholesterol and atherosclerosis in patients with type 2 diabetes mellitus: a randomised controlled trial",
      "authors": "Brousseau ME, Schaefer EJ, Wolfe ML, et al.",
      "publication_name": "New England Journal of Medicine",
      "publication_date": "2004-06-03",
      "doi_or_url": "10.1056/NEJMoa031766",
      "retrieval_date": "2026-06-10",
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      "admission_decision": "ADMITTED",
      "evidence_type": "Phase II efficacy data — HDL-C increase, LDL-C reduction",
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      "authors": "Brousseau ME, Schaefer EJ, Wolfe ML, et al.",
      "publication_name": "New England Journal of Medicine",
      "publication_date": "2004-06-03",
      "doi_or_url": "10.1056/NEJMoa031766",
      "retrieval_date": "2026-06-10",
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      "admission_decision": "ADMITTED",
      "evidence_type": "Phase II efficacy data — HDL-C increase, LDL-C reduction",
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      "doc_type": "JOURNAL_ARTICLE",
      "title": "Effects of torcetrapib on HDL cholesterol and atherosclerosis in patients with type 2 diabetes mellitus: a randomised controlled trial",
      "authors": "Brousseau ME, Schaefer EJ, Wolfe ML, et al.",
      "publication_name": "New England Journal of Medicine",
      "publication_date": "2004-06-03",
      "doi_or_url": "10.1056/NEJMoa031766",
      "retrieval_date": "2026-06-10",
      "admissibility_category": "A",
      "admission_decision": "ADMITTED",
      "evidence_type": "Phase II efficacy data — HDL-C increase, LDL-C reduction",
      "key_data_points": "HDL-C increase 46-106% dose-dependent; LDL-C reduction 50-65% dose-dependent",
      "leakage_risk_flag": false
    },
    {
      "doc_id": "DOC-ATM-2006-001-007",
      "doc_type": "JOURNAL_ARTICLE",
      "title": "Effects of torcetrapib on HDL cholesterol and atherosclerosis in patients with type 2 diabetes mellitus: a randomised controlled trial",
      "authors": "Brousseau ME, Schaefer EJ, Wolfe ML, et al.",
      "publication_name": "New England Journal of Medicine",
      "publication_date": "2004-06-03",
      "doi_or_url": "10.1056/NEJMoa031766",
      "retrieval_date": "2026-06-10",
      "admissibility_category": "A",
      "admission_decision": "ADMITTED",
      "evidence_type": "Phase II efficacy data — HDL-C increase, LDL-C reduction",
      "key_data_points": "HDL-C increase 46-106% dose-dependent; LDL-C reduction 50-65% dose-dependent",
      "leakage_risk_flag": false
    },
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      "doc_id": "DOC-ATM-2006-001-008",
      "doc_type": "JOURNAL_ARTICLE",
      "title": "Effects of torcetrapib on HDL cholesterol and atherosclerosis in patients with type 2 diabetes mellitus: a randomised controlled trial",
      "authors": "Brousseau ME, Schaefer EJ, Wolfe ML, et al.",
      "publication_name": "New England Journal of Medicine",
      "publication_date": "2004-06-03",
      "doi_or_url": "10.1056/NEJMoa031766",
      "retrieval_date": "2026-06-10",
      "admissibility_category": "A",
      "admission_decision": "ADMITTED",
      "evidence_type": "Phase II efficacy data — HDL-C increase, LDL-C reduction",
      "key_data_points": "HDL-C increase 46-106% dose-dependent; LDL-C reduction 50-65% dose-dependent",
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}
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"doc_id": "DOC-ATM-2006-001-002",
"doc_type": "JOURNAL_ARTICLE",
"title": "CETP inhibition: from bench to bedside",
"authors": "Tall AR, Yvan-Charvet L, Wang N",
"publication_name": "Arteriosclerosis, Thrombosis, and Vascular Bio",
"publication_date": "2007-02-01",
"doi_or_url": "10.1161/01.ATV.0000261081.02375.d0",
"retrieval_date": "2026-06-10",
"admissibility_category": "B",
"admission_decision": "EXCLUDED",
"exclusion_category": "POST_CUTOFF",
"exclusion_rule": "Rule 1.1 – Published after Evidence Cutoff Date",
"curator_decision": "Published February 2007, 28 months after evidence"
}
]
}
```

PART 4 — Leakage Prevention Protocol

Leakage Pathway Classification

Evidence leakage occurs when information about the Phase III outcome influences the council's deliberation, either directly (explicit outcome reference) or indirectly (outcome-informed framing). The following matrix identifies all known leakage pathways, ranked by severity.

Leakage Prevention Matrix

Rank	Pathway	Severity	Detection Method	Mitigation Strategy
1	LLM latent knowledge	CRITICAL	Undetectable by architecture alone	RAG-only execution model; disclosed limitation
2	Training-data contamination	CRITICAL	Undetectable	Model version locking; disclosed limitation
3	Direct outcome reference in evidence corpus	CRITICAL	Automated date-check + manual review	Evidence Manifest + Evidence Curator review
4	Retrospective review articles	HIGH	Publication date check	Category B exclusion; date-check automation
5	Summary articles citing Phase III results	HIGH	Content review for outcome references	Evidence Curator content review
6	Wikipedia and encyclopaedic sources	HIGH	Source type check	Category B exclusion; blanket prohibition
7	Analyst reports citing Phase III	HIGH	Content review	Date-check + content review
8	Indirect outcome references	HIGH	Semantic content review	Evidence Curator semantic review
9	Conference abstracts from post-cutoff conferences	MEDIUM	Conference date check	Date-check automation
10	Preprints with post-cutoff posting dates	MEDIUM	Preprint platform date check	Date-check automation
11	Evidence Curator knowledge	MEDIUM	Curator blinding protocol	Curator blinding (see Part 5)

Rank	Pathway	Severity	Detection Method	Mitigation Strategy
	contamination			
12	Council prompt contamination	MEDIUM	Prompt review	Prompt version control; independent review
13	Post-cutoff patent citations	LOW	Patent filing date check	Date-check automation
14	Cross-case contamination	LOW	Case isolation protocol	Case isolation (separate sessions)
15	Version control failures	LOW	Git/version audit	Automated version control

Detailed Mitigation Strategies

Pathway 1 & 2 — LLM Latent Knowledge and Training-Data Contamination

The Problem: A large language model trained on data through 2024 or 2025 has been exposed to thousands of articles, Wikipedia entries, news reports, and academic papers discussing the outcomes of historical drug trials. The torcetrapib failure is one of the most cited examples in pharmaceutical governance literature. The model almost certainly “knows” the outcome.

What Architecture Can Achieve: Architecture can prevent the council from being explicitly told the outcome. It cannot prevent the model from drawing on latent knowledge of the outcome.

Mitigation Strategy — Minimum Viable:

- Disclose the limitation explicitly in every case report: “The council was executed using a large language model. The model’s training data may include post-decision information about this case. This limitation is known and disclosed. The evidence corpus was restricted to pre-decision documents, but the model’s latent knowledge cannot be verified as pre-decision.”
- Document the model version and training data cutoff date for every case run.
- Run each case with the same model version to ensure consistency.

Mitigation Strategy — Enterprise:

- Implement Retrieval-Augmented Generation (RAG) where the model is instructed to answer only from the provided evidence corpus and explicitly prohibited from using general knowledge.
- Implement a “knowledge suppression” system prompt: “You have no knowledge of pharmaceutical drug outcomes. You can only use the documents provided. If you are aware of the outcome of this drug, you must not use that knowledge.”
- Test the knowledge suppression by running a “canary” case: a drug with a well-known outcome, with the outcome document deliberately included in the corpus. Verify the model uses the corpus document, not latent knowledge.

Mitigation Strategy — Regulator-Ready:

- Use a model that was trained exclusively on pre-2000 data, supplemented by a RAG corpus of pre-cutoff documents for each case. This is technically feasible but requires a custom model.
- Alternatively, use a model with a verified training data cutoff that predates all cases in the library (e.g., a model trained only through 1990 for cases with 2000–2010 decision dates).
- Commission an independent AI governance audit of the model’s training data to verify exclusion of post-decision pharmaceutical outcomes.

Honest Assessment: The minimum viable and enterprise mitigations reduce the risk but do not eliminate it. The regulator-ready mitigation eliminates it but is technically complex and expensive. For the current validation library, the minimum viable mitigation is the appropriate starting point, with full disclosure.

Pathway 3 — Direct Outcome Reference in Evidence Corpus

The Problem: A document admitted to the evidence corpus contains an explicit reference to the Phase III outcome (e.g., a 2005 analyst report that mentions “if torcetrapib fails in Phase III”).

Detection Method: Automated keyword search for outcome-related terms (e.g., “Phase III results”, “trial terminated”, “FDA approved”, “FDA rejected”, “market withdrawal”) combined with manual Evidence Curator review.

Mitigation: Automated pre-screening of all documents for outcome-related keywords. Documents flagged by the automated screen are reviewed by the Evidence Curator before admission. Any document containing an explicit outcome reference is excluded as Category B evidence.

Pathway 4 & 5 — Retrospective Review Articles and Summary Articles

The Problem: A review article published after the Evidence Cutoff Date summarises the Phase II data in the context of the Phase III outcome. Even if the review article is excluded from the corpus, the Evidence Curator may have read it and been influenced by its framing.

Detection Method: Publication date check (automated). Content review for outcome-informed framing (manual).

Mitigation: Automated exclusion of all documents published after the Evidence Cutoff Date. Evidence Curator blinding protocol (see Part 5).

Pathway 8 — Indirect Outcome References

The Problem: A document admitted to the corpus does not explicitly mention the Phase III outcome but frames the Phase II data in a way that implies knowledge of the outcome. Example: “The blood pressure increase observed in Phase II was later confirmed to be a mechanism-based effect” — this sentence implies knowledge of the post-Phase III confirmation.

Detection Method: Semantic content review by Evidence Curator. Automated keyword search for temporal markers (“later”, “subsequently”, “was confirmed”, “proved to be”).

Mitigation: Evidence Curator semantic review of all admitted documents. Automated flagging of temporal markers for manual review.

Pathway 11 — Evidence Curator Knowledge Contamination

The Problem: The Evidence Curator knows the outcome of the case and unconsciously selects documents that support the correct verdict, or frames the evidence summary in outcome-informed language.

Detection Method: Curator blinding protocol (see Part 5). Independent review of evidence summary by a blinded reviewer.

Mitigation: The Evidence Curator is blinded to the Phase III outcome before completing the evidence manifest. The outcome is revealed only after the council run is complete and the manifest is locked.

PART 5 — Blinded Case Methodology

Blinding Protocol

The blinded case methodology ensures that the council’s deliberation is conducted without knowledge of the Phase III outcome, and that the evidence package cannot be altered after the council run.

Workflow

PHASE 1 – CASE SELECTION

- Case Selector identifies candidate case
- Case Selector records: Drug, Sponsor, Indication, Approximate Decision Year
- Case Selector does NOT record Phase III outcome at this stage

PHASE 2 – EVIDENCE CURATION (BLINDED)

- Evidence Curator receives: Drug, Sponsor, Indication, Decision Year
- Evidence Curator does NOT receive: Phase III outcome
- Evidence Curator builds Evidence Corpus:
 - Searches PubMed, ClinicalTrials.gov, FDA.gov, EMA.europa.eu
 - Applies Evidence Cutoff Date (Decision Date - 30 days)
 - Assigns Document IDs
 - Completes Evidence Manifest
 - Signs Evidence Manifest (digital signature)
- Evidence Corpus is LOCKED (hash recorded)
- Evidence Curator is BLINDED to Phase III outcome throughout

PHASE 3 – COUNCIL EXECUTION

- Council receives ONLY the Evidence Corpus (no additional context)
- Council session ID is generated and recorded
- Council deliberation is executed
- Council output is recorded (votes, rationales, verdict)
- Council output is LOCKED (hash recorded)
- Council output is NOT reviewed before Phase 4

PHASE 4 – OUTCOME REVELATION

- Phase III outcome is retrieved from public sources
- Outcome is recorded in Retrospective Outcome Record
- Outcome is compared to Council verdict
- Comparison result is recorded

PHASE 5 – VALIDATION SCORING

- Independent Reviewer receives: Evidence Corpus, Council Output, Outcome
- Independent Reviewer scores: Verdict Alignment, OVS, DDIA
- Independent Reviewer signs Validation Scorecard
- Validation Scorecard is LOCKED

PHASE 6 – AUDIT TRAIL GENERATION

- All records are compiled into Case Audit Package
- Case Audit Package includes:
 - Evidence Manifest (signed, locked)

- Council Output (signed, locked)
 - Retrospective Outcome Record
 - Validation Scorecard (signed, locked)
 - Hash verification for all locked records
- Case Audit Package is archived

Governance Checkpoints

Checkpoint	Trigger	Required Action	Responsible Party
CP-1	Evidence Corpus completed	Evidence Curator signs manifest	Evidence Curator
CP-2	Evidence Corpus locked	Hash recorded; no further changes permitted	System
CP-3	Council execution complete	Council output locked	System
CP-4	Outcome revealed	Outcome recorded; comparison completed	Validation Director
CP-5	Validation scoring complete	Independent Reviewer signs scorecard	Independent Reviewer
CP-6	Case Audit Package complete	Archived; hash verified	Governance Office

Immutability Controls

Evidence Corpus Lock: The evidence corpus is locked (SHA-256 hash recorded) before the council is executed. Any subsequent modification to the corpus invalidates the case.

Council Output Lock: The council output is locked (SHA-256 hash recorded) immediately after execution. Any subsequent modification to the output invalidates the case.

Manifest Lock: The evidence manifest is locked after Evidence Curator sign-off. Any subsequent modification requires a new manifest version and a new council run.

Audit Trail: All lock events are recorded in the Governance Audit Log with timestamp, user ID, and hash value.

PART 6 — Reproducibility Standard

The Reproducibility Problem

An independent party must be able to:

1. Access the same evidence package
2. Rerun the council with the same configuration
3. Compare outputs
4. Reach the same conclusion about verdict alignment

This requires full publication of the methodology, the evidence corpus, the constitutional rules, the persona definitions, and the exact system configuration.

Step-by-Step Reproducibility Framework

Step 1 — Access the Evidence Package

The Evidence Package for each case is published in a public repository (GitHub or equivalent) containing:

- Evidence Manifest (JSON, signed)
- All admitted documents (PDF or DOI links)
- Evidence Corpus Summary (structured text)
- Exclusion Log (JSON)
- Hash values for all locked records

Access requirement: Any party with internet access can download the Evidence Package. No proprietary access required.

Step 2 — Verify the Evidence Boundary

The independent party verifies:

- Decision Date (check against published sources)
- Evidence Cutoff Date (Decision Date – 30 days)
- All admitted documents have publication dates before the Evidence Cutoff Date
- All excluded documents have correct exclusion rationales

Verification tool: Automated date-check script (Python, open source, published in repository).

Step 3 — Access the Methodology

The Methodology Document is published and includes:

- Pharma Constitution version used (full text)
- Persona definitions (full text)
- Decision Brief template (full text)
- System prompt (full text)
- Model name and version
- Temperature and other generation parameters
- Council execution protocol

Access requirement: Public repository.

Step 4 — Rerun the Council

The independent party executes the council using:

- The published Evidence Corpus Summary as input
- The published Methodology Document as configuration
- The same model version (or a documented alternative if the original is unavailable)

Expected output: The independent party's council output will not be identical to the original (LLM outputs are non-deterministic) but should produce the same verdict in $\geq 80\%$ of reruns. This is the **Reproducibility Threshold**.

Reproducibility Threshold: A case is considered reproducible if an independent party running the council with the published methodology produces the same verdict (GO / WAIT / NO-GO) in $\geq 80\%$ of 10 independent reruns.

Step 5 — Compare Outputs

The independent party compares:

- Verdict (GO / WAIT / NO-GO)
- Primary blockers identified
- Constitutional rules triggered
- Confidence levels

Comparison metric: Verdict agreement rate across 10 reruns. Primary blocker overlap (Jaccard similarity).

Step 6 — Report Reproducibility Finding

The independent party publishes:

- Verdict agreement rate
- Primary blocker overlap
- Any systematic differences between original and reproduced outputs
- Conclusion: REPRODUCIBLE / PARTIALLY REPRODUCIBLE / NOT REPRODUCIBLE

Reproducibility thresholds:

- REPRODUCIBLE: ≥80% verdict agreement across 10 reruns
 - PARTIALLY REPRODUCIBLE: 60–79% verdict agreement
 - NOT REPRODUCIBLE: <60% verdict agreement
-

PART 7 — Independent Review Protocol

FDA Reviewer

Profile: Former FDA reviewer with experience in pharmaceutical governance and clinical trial methodology. Focused on GxP compliance, audit trail integrity, and evidence quality.

Primary Objections:

1. “The evidence boundary is a prompt instruction, not a verifiable constraint.”
2. “The LLM training data may include post-decision information.”
3. “The audit trail is not 21 CFR Part 11 compliant.”
4. “The methodology is not reproducible.”

Required Evidence:

- Full Evidence Manifest for each case (signed, locked, with hash values)
- Published methodology document (model version, temperature, system prompt)
- Reproducibility data (inter-run consistency across 10 reruns)
- Disclosure of LLM training data limitation

- 21 CFR Part 11 compliance assessment

Acceptance Threshold: The FDA will not formally accept or endorse this methodology for regulatory use without a prospective validation study. For informal review, the FDA will accept the methodology as “exploratory” if the evidence boundary limitation is fully disclosed and the audit trail is complete.

Recommended Documentation:

- Evidence Boundary Architecture document (this document)
- Case Audit Packages for all completed cases
- Reproducibility study report
- LLM training data limitation disclosure

EMA Reviewer

Profile: Former EMA reviewer with experience in scientific advice and methodology review. Focused on transparency, reproducibility, and methodology documentation.

Primary Objections:

1. “The methodology is not published in a peer-reviewed journal.”
2. “The reproducibility has not been demonstrated.”
3. “The constitutional rules have not been validated against a pre-specified case set.”

Required Evidence:

- Published methodology paper (or preprint)
- Reproducibility data
- Pre-specified case selection criteria
- Constitutional rule validation data

Acceptance Threshold: The EMA will accept the methodology for informal review if it is published (preprint acceptable) and reproducibility data is provided. Formal acceptance requires peer-reviewed publication.

Recommended Documentation:

- Methodology preprint (bioRxiv or SSRN)
- Reproducibility study report
- Pre-specified case selection criteria document

Top-20 Pharmaceutical Company

Profile: Head of R&D or Chief Medical Officer. Focused on practical value-add over existing processes.

Primary Objections:

1. "What does this catch that our internal teams miss?"
2. "Show me a prospective example."
3. "What is the liability if your council is wrong?"

Required Evidence:

- Torcetrapib case study (the clearest value-add example)
- Prospective pilot proposal
- Liability framework

Acceptance Threshold: The pharma company will engage in a pilot discussion if presented with the torcetrapib case and a clear prospective pilot design. They will not pay for a service based solely on retrospective validation.

Recommended Documentation:

- Torcetrapib Institutional Proof Report
- Prospective pilot proposal (blinded, 5 current Phase II drugs)
- Liability and confidentiality framework

CRO (IQVIA, ICON, Covance)

Profile: Head of Regulatory Affairs or Scientific Advisory. Focused on integration with existing trial management processes.

Primary Objections:

1. "How does this integrate with our existing governance processes?"
2. "Who is responsible for the council output?"
3. "Is this compatible with ICH E6(R2) GCP requirements?"

Required Evidence:

- Integration protocol with existing Phase II review processes
- Responsibility matrix (who owns the council output)

- ICH E6(R2) compatibility assessment

Acceptance Threshold: The CRO will engage if the methodology can be positioned as a supplementary governance layer, not a replacement for existing processes.

Recommended Documentation:

- Integration protocol
- Responsibility matrix
- ICH E6(R2) compatibility assessment

Big Four Advisory Team

Profile: Life sciences diligence partner. Focused on methodology defensibility in an IC memo.

Primary Objections:

1. “Can I cite this in a diligence report?”
2. “Is the methodology independently verifiable?”
3. “What are the disclosed limitations?”

Required Evidence:

- Published methodology (peer-reviewed or preprint)
- Full disclosure of limitations
- Independent expert validation of at least one case

Acceptance Threshold: The Big Four will cite the methodology in a diligence report if it is published, limitations are fully disclosed, and at least one case has been independently validated.

Recommended Documentation:

- Published methodology paper
 - Limitations disclosure document
 - Independent expert validation report (torcetrapib)
-

PART 8 — Validation Library Admission Rules

Admission Criteria

A historical case may be admitted to the Validation Library if and only if it satisfies all of the following criteria:

Criterion 1 — Minimum Evidence Availability

The case must have at least **three** Category A documents available in the public domain, covering:

- Phase II efficacy data (at least one published trial result)
- Phase II safety data (at least one safety report or adverse event summary)
- Regulatory context (at least one regulatory guidance document or advisory committee transcript)

Minimum Evidence Score: 3 Category A documents. Cases with fewer than 3 documents are classified as **Evidence-Insufficient (EI)** and excluded.

Criterion 2 — Minimum Documentation Quality

All admitted documents must be:

- Primary sources (peer-reviewed publications, regulatory documents, clinical trial registry entries)
- Publicly accessible (DOI or URL verifiable)
- Date-verified (publication date confirmed independently)

Documentation Quality Score: 100% of admitted documents must satisfy all three quality criteria.

Criterion 3 — Known Outcome

The Phase III outcome must be publicly documented and unambiguous. Cases where the Phase III outcome is disputed or unknown are excluded.

Known Outcome Types: Successful approval / Failed Phase III / Withdrawn / Regulatory Controversy (with documented basis) / Terminated for futility / Terminated for safety

Criterion 4 — Decision Traceability

The Phase II → Phase III advancement decision must be traceable to a verifiable source (press release, ClinicalTrials.gov registration, IND filing date, or equivalent). Cases where the decision date cannot be established within a 180-day window are classified as **Decision Date Uncertain (DDU)** and excluded until the date is resolved.

Criterion 5 — Therapeutic Area Coverage

The case must be in one of the pre-specified therapeutic areas for the current library expansion. Cases outside the pre-specified areas are deferred to future library expansions.

Criterion 6 — No Active Litigation

Cases where the Phase III outcome or regulatory decision is the subject of active litigation are excluded until the litigation is resolved.

Exclusion Criteria

A case is excluded from the Validation Library if any of the following apply:

Exclusion Criterion	Rationale
Fewer than 3 Category A documents	Insufficient evidence for meaningful council deliberation
Decision date cannot be established within 180 days	Evidence boundary cannot be reliably defined
Phase III outcome is disputed or unknown	Verdict alignment cannot be assessed
Active litigation involving the Phase III outcome	Legal risk; outcome may change
Case involves a drug currently in active development	Outcome contamination risk; prospective, not retrospective
Evidence Boundary Violation detected and not resolved	Case integrity compromised
Case was selected after Phase III outcome was known to the case selector	Selection bias; excluded unless pre-specification criteria are documented

Conditional Admission

Cases that satisfy Criteria 1–4 but have one or more of the following characteristics are admitted with a **Conditional Admission Flag (CAF)**:

Condition	CAF Code	Disclosure Required
Decision date confidence is MEDIUM or LOW	CAF-01	Disclose uncertainty range
Fewer than 5 Category A documents	CAF-02	Disclose evidence limitation
Phase III outcome is Regulatory Controversy	CAF-03	Disclose outcome ambiguity
Case involves accelerated approval pathway	CAF-04	Disclose pathway-specific considerations
Case involves ultra-rare disease (n<100 in pivotal)	CAF-05	Disclose evidence threshold limitations

PART 9 — Governance and Audit Layer

Governance Structure

The Validation Library governance structure consists of four roles:

Role	Responsibilities	Independence Requirement
Validation Director	Overall library governance; case selection; final approval	Must not be involved in council execution
Evidence Curator	Evidence corpus construction; manifest completion; blinding protocol	Must be blinded to Phase III outcome during curation
Council Operator	Council execution; session management; output locking	Must not be involved in evidence curation
Independent Reviewer	Validation scoring; DDIA assessment; reproducibility testing	Must be external to AgenThink Mesh

Audit Log Schema

Every action in the Validation Library is recorded in the Governance Audit Log.

Field	Type	Description
log_id	String	Unique log entry identifier
timestamp	DateTime	ISO 8601 with timezone
actor_id	String	User or system ID
actor_role	Enum	VALIDATION_DIRECTOR / EVIDENCE_CURATOR / COUNCIL_OPERATOR / INDEPENDENT_REVIEWER / SYSTEM
action_type	Enum	CASE_CREATED / DOCUMENT_ADMITTED / DOCUMENT_EXCLUDED / CORPUS_LOCKED / COUNCIL_EXECUTED / OUTPUT_LOCKED / OUTCOME_RECORDED / SCORE_RECORDED / AMENDMENT_PROPOSED / AMENDMENT_RATIFIED / EBV_DETECTED / EBV_RESOLVED
case_id	String	Associated case ID
doc_id	String	Associated document ID (if applicable)
action_detail	String	Free-text description of action
hash_before	String	SHA-256 hash of affected record before action
hash_after	String	SHA-256 hash of affected record after action
signature	String	Digital signature of actor

Chain of Custody

The chain of custody for each case follows this sequence:

Case Created → Evidence Corpus Built → Corpus Locked → Council Executed → Output Locked → Outcome Revealed → Validation Scored → Audit Package Comp. → Audit Package Archived

Each step is recorded in the Governance Audit Log with timestamp, actor, and hash values. The chain is unbroken if every step has a corresponding log entry with valid hash values.

Version Control

Constitutional Version Tracking:

- Each version of the Pharma Constitution is stored in a version-controlled repository (Git)
- Each case is tagged with the constitutional version used
- Amendments are tracked with: amendment ID, rationale, cases affected, ratification date, ratifying authority

Evidence Package Versioning:

- Each Evidence Manifest has a semantic version (major.minor.patch)
- Major version: new documents admitted or excluded
- Minor version: metadata corrections
- Patch version: formatting corrections
- Version history is preserved; no version is deleted

Council Output Versioning:

- Each council output is immutable after locking
- If a case requires re-execution (e.g., after an Evidence Boundary Violation is resolved), a new council session is created with a new session ID
- The original output is preserved; the new output is tagged as a revision

Amendment Tracking

Field	Description
<code>amendment_id</code>	Unique identifier (format: PC-[VERSION]-[SEQ])
<code>proposed_by</code>	Proposing party
<code>proposed_date</code>	ISO 8601
<code>trigger_case</code>	Case ID that triggered the amendment
<code>rule_text</code>	Full text of proposed rule
<code>rationale</code>	Detailed rationale
<code>cases_affected</code>	List of case IDs where rule would have applied
<code>validation_cases</code>	List of case IDs used to validate the amendment
<code>validation_result</code>	VALIDATED / NOT_VALIDATED / PENDING
<code>ratification_date</code>	ISO 8601 (if ratified)
<code>ratifying_authority</code>	Name and role
<code>status</code>	PROPOSED / VALIDATED / RATIFIED / REJECTED

PART 10 — Gold Standard Architecture

Three Architecture Tiers

Tier 1 — Minimum Viable Architecture (60–90 days)

What it achieves: Defensible evidence boundary with disclosed limitations. Sufficient for internal demonstration and early pilot discussions.

Components:

- Evidence Manifest System (manual, structured JSON)
- Evidence Cutoff Date governance (Decision Date – 30 days)
- Blinded Case Methodology (curator blinding; corpus locking)

- Governance Audit Log (manual entry)
- Published methodology document (model version, temperature, prompts)
- LLM training data limitation disclosure

What it does not achieve:

- Verifiable LLM knowledge exclusion
- Automated date-check and leakage detection
- Reproducibility testing
- Independent expert validation

Cost estimate: 2–3 FTE for 60–90 days. No new technology required.

Reviewer verdict: PARTIALLY ACCEPTABLE to FDA/EMA. Acceptable for internal demonstration. Not sufficient for enterprise pilots without additional disclosures.

Tier 2 — Enterprise Architecture (6–12 months)

What it achieves: Verifiable evidence boundary with automated controls. Sufficient for enterprise pilots and peer-review submission.

Components (all Tier 1 components, plus):

- Automated date-check pipeline (Python; PubMed API; ClinicalTrials.gov API)
- Automated leakage detection (keyword search; temporal marker flagging)
- Retrieval-Augmented Generation (RAG) execution model
- Knowledge suppression system prompt with canary testing
- Reproducibility testing protocol (10 reruns per case)
- Independent expert validation (3 external reviewers per case)
- Public evidence corpus repository (GitHub)
- 21 CFR Part 11 compliant audit log (electronic signatures; timestamp authority)

What it does not achieve:

- Fully verifiable LLM knowledge exclusion (RAG reduces but does not eliminate latent knowledge risk)
- Regulatory submission use
- Prospective validation track record

Cost estimate: 5–8 FTE for 6–12 months. Technology investment: 200K–500K (RAG infrastructure, audit log system, repository).

Reviewer verdict: ACCEPTABLE to FDA/EMA for informal review. Sufficient for enterprise pilot discussions. Peer-review eligible.

Tier 3 — Regulator-Ready Architecture (18–24 months)

What it achieves: Fully verifiable evidence boundary. Sufficient for regulatory submission consideration and peer-reviewed publication.

Components (all Tier 2 components, plus):

- Custom model with verified training data cutoff (pre-dates all library cases)
- OR: Model with verified pharmaceutical outcome exclusion from training data
- Formal FDA/EMA pre-submission meeting
- Prospective validation pilot (blinded, 5–10 current Phase II drugs)
- Peer-reviewed methodology paper (published)
- Independent AI governance audit (third-party)
- Liability framework (legal opinion on advisory liability)
- GxP-compliant quality management system

What it does not achieve:

- Regulatory approval for clinical decision-making use (requires prospective validation at scale)

Cost estimate: 10–15 FTE for 18–24 months. Technology investment: 1M–3M (custom model or model audit, QMS, prospective pilot infrastructure).

Reviewer verdict: ACCEPTABLE to FDA/EMA for formal review. Sufficient for peer-reviewed publication. Sufficient for enterprise deployment with disclosed limitations.

Architecture Comparison

Dimension	Tier 1 (MVP)	Tier 2 (Enterprise)	Tier 3 (Regulator-Ready)
Evidence boundary verifiability	Partial (manual)	High (automated)	Full (verified model)
LLM knowledge exclusion	Disclosed limitation	Partially mitigated (RAG)	Fully mitigated (custom model)
Audit trail completeness	Manual	21 CFR Part 11 compliant	GxP compliant
Reproducibility	Not tested	Tested (10 reruns)	Tested + published
Independent validation	None	3 experts per case	Peer-reviewed
FDA/EMA acceptance	Informal only	Informal review	Formal review eligible
Enterprise pilot eligibility	With disclosure	Yes	Yes
Peer-review eligibility	No	Yes	Yes
Timeline	60–90 days	6–12 months	18–24 months
Cost estimate	50K–100K	500K–1M	2M–5M

DELIVERABLE 1 — Full Architecture Summary

The Evidence Boundary Architecture consists of ten integrated components:

- Evidence Admissibility Standard** — formal rules governing which documents may enter the council’s evidence corpus, with three categories (Admissible, Prohibited, Conditional) and specific verification methods for each document type.
- Time Cutoff Governance** — rules for establishing the Decision Date and Evidence Cutoff Date, with priority order for source selection and specific governance rules for edge cases (embargoes, retrospective registrations, rolling publications).

3. **Evidence Manifest System** — a structured JSON schema recording every document considered for council input, with case-level and document-level fields, digital signatures, and hash-based immutability controls.
 4. **Leakage Prevention Protocol** — a 15-pathway leakage matrix ranked by severity, with detection methods and mitigation strategies for each pathway, including the honest acknowledgement that LLM latent knowledge cannot be fully mitigated by architecture alone.
 5. **Blinded Case Methodology** — a six-phase workflow with governance checkpoints, immutability controls, and role separation ensuring the council cannot see the outcome and the evidence package cannot be altered after execution.
 6. **Reproducibility Standard** — a six-step framework enabling independent parties to access the same evidence package, rerun the council, and compare outputs, with a defined Reproducibility Threshold ($\geq 80\%$ verdict agreement across 10 reruns).
 7. **Independent Review Protocol** — reviewer-specific objections, required evidence, acceptance thresholds, and recommended documentation for FDA, EMA, top-20 pharma, CRO, and Big Four reviewers.
 8. **Validation Library Admission Rules** — six admission criteria and eight exclusion criteria governing case entry, with a Conditional Admission Flag system for cases with disclosed limitations.
 9. **Governance and Audit Layer** — four-role governance structure, Governance Audit Log schema, chain-of-custody protocol, version control for constitutional amendments and evidence packages, and 21 CFR Part 11 compliance pathway.
 10. **Gold Standard Architecture** — three implementation tiers (Minimum Viable, Enterprise, Regulator-Ready) with cost estimates, timelines, and reviewer acceptance verdicts for each.
-

DELIVERABLE 8 — Gap Analysis of AgenThink's Current Approach

Gap	Current State	Required State	Severity	Priority
Evidence boundary verifiability	Prompt instruction only	Evidence Manifest + automated date-check	CRITICAL	P1
LLM knowledge exclusion	Undisclosed limitation	Disclosed limitation + RAG (Tier 2)	CRITICAL	P1
Audit trail	None	Governance Audit Log with hash values	HIGH	P1
Methodology publication	Unpublished	Published (preprint minimum)	HIGH	P1
Reproducibility testing	Not performed	10 reruns per case	HIGH	P2
Independent expert validation	None	3 experts per case (Tier 2)	HIGH	P2
Case selection pre-specification	Retrospective	Pre-specified criteria before case selection	HIGH	P2
Constitutional amendment freeze	Not implemented	5-case validation test before ratification	MEDIUM	P2
21 CFR Part 11 compliance	Not assessed	Compliance assessment (Tier 2)	MEDIUM	P3
Prospective validation	Not started	Blinded pilot (5–10 current Phase II drugs)	MEDIUM	P3
Liability framework	None	Legal opinion on advisory liability	MEDIUM	P3
GxP compliance	Not assessed	GxP assessment (Tier 3)	LOW	P4

DELIVERABLE 9 — Prioritised Implementation Roadmap

Phase 1 — Foundation (Weeks 1–8): Minimum Viable Architecture

Week 1–2:

- Draft and publish the Evidence Boundary Architecture document (this document)
- Disclose the LLM training data limitation in all existing case reports
- Document the model version and temperature used for all 10 completed cases

Week 3–4:

- Build the Evidence Manifest template (JSON schema)
- Retroactively complete Evidence Manifests for all 10 completed cases
- Identify and document all documents used in each case

Week 5–6:

- Implement the curator blinding protocol for future cases
- Implement corpus locking (SHA-256 hash) for future cases
- Implement council output locking for future cases

Week 7–8:

- Publish the methodology document (model version, temperature, system prompt, constitutional rules, persona definitions)
- Create the public evidence corpus repository (GitHub)
- Publish Evidence Manifests for all 10 completed cases

Milestone: Minimum Viable Architecture complete. Cases 1–10 retroactively documented. Ready for early pilot discussions with full disclosure.

Phase 2 — Automation (Months 3–6): Enterprise Architecture

Month 3:

- Build automated date-check pipeline (PubMed API, ClinicalTrials.gov API)
- Build automated leakage detection (keyword search, temporal marker flagging)
- Implement Governance Audit Log

Month 4:

- Implement RAG execution model
- Implement knowledge suppression system prompt
- Run canary test cases to validate knowledge suppression

Month 5:

- Run reproducibility testing for all 10 completed cases (10 reruns each)
- Publish reproducibility data
- Commission 3 independent pharmaceutical experts to validate DDIA for 5 cases

Month 6:

- Submit methodology preprint (bioRxiv or SSRN)
- Pre-specify case selection criteria for Cases 11–50
- Begin Cases 11–20 (Oncology batch) with full Enterprise Architecture

Milestone: Enterprise Architecture complete. Peer-review submission ready. Enterprise pilot discussions can begin.

Phase 3 — Validation (Months 7–18): Regulator-Ready Architecture**Month 7–9:**

- Complete Cases 11–30 with Enterprise Architecture
- Publish interim results (20-case library)
- Design prospective pilot (5 current Phase II drugs, blinded)

Month 10–12:

- Complete Cases 31–50 with Enterprise Architecture
- Submit peer-reviewed methodology paper
- Launch prospective pilot

Month 13–18:

- Complete peer-review process
- Engage FDA/EMA for informal methodology review

- Engage legal counsel on advisory liability framework
- Assess GxP compliance requirements
- Prepare Regulator-Ready Architecture implementation plan

Milestone: Regulator-Ready Architecture implementation plan complete. Peer-reviewed publication submitted. Prospective pilot underway.

DELIVERABLE 10 — Executive Verdict

The Final Answer

Would this architecture satisfy a skeptical FDA reviewer who believes the Council may have known the answer?

PARTIALLY — with a clear path to YES.

At Tier 1 (Minimum Viable): PARTIALLY. The architecture provides a documented evidence boundary, a disclosed limitation, and an audit trail. The FDA reviewer will acknowledge the effort but will note: “You’ve documented what you gave the model. You haven’t proven the model didn’t already know the answer.” This is correct. The honest response is: “You’re right. We can’t prove it. We’ve disclosed it. Here’s what we’re doing about it.”

At Tier 2 (Enterprise): PARTIALLY — approaching YES. The RAG execution model reduces (but does not eliminate) the latent knowledge risk. The reproducibility data demonstrates methodological consistency. The independent expert validation provides a human-readable audit trail. The FDA reviewer will say: “This is better. The RAG architecture is a meaningful control. But I still can’t verify the model’s training data.” The honest response is: “Correct. We’re working on it.”

At Tier 3 (Regulator-Ready): YES — for informal review. The custom model with verified training data cutoff eliminates the latent knowledge objection. The prospective pilot provides a forward-looking track record. The peer-reviewed publication provides independent methodology validation. The FDA reviewer will say: “This is defensible for exploratory use. I’m not endorsing it for regulatory decision-making, but I’m not objecting to it either.”

The Three Things That Must Be True

For this architecture to satisfy a skeptical FDA reviewer:

1. **The LLM training data limitation must be disclosed, not hidden.** Every case report, every presentation, every pilot proposal must include the sentence: “The council was executed using a large language model. The model’s training data may include post-decision information about this case. This limitation is known, disclosed, and partially mitigated by [specific mitigation].”
2. **The evidence corpus must be independently verifiable.** Every document in the corpus must have a DOI or URL, a verified publication date, and a documented admission decision. An independent party must be able to download every document and verify that it was published before the Evidence Cutoff Date.
3. **The methodology must be published.** The exact prompts, model version, constitutional rules, and persona definitions must be in the public domain before any enterprise discussion. A methodology that cannot be independently replicated is not a methodology — it is a black box.

The Single Most Important Action

Build the Evidence Corpus before running any more cases.

Every case run without a verified evidence corpus is a case that cannot be defended. The 10 completed cases are currently defended by a prompt instruction. That is not sufficient. The retroactive Evidence Manifests (Phase 1, Weeks 3–4) will partially address this. The automated date-check pipeline (Phase 2, Month 3) will fully address it.

The architecture described in this document is implementable. The timeline is realistic. The cost is proportionate to the commercial opportunity. The honest limitation is known and manageable.

Run the architecture. Then run the cases.

End of Document

AgenThink Mesh Evidence Boundary Architecture v1.0 | June 2026

Classification: Governance Document — Internal & External Review